19 FEBRERO DE 2015

1) $y[n] = \sum_{k=-\infty}^{n-1} \alpha^{n-k} \{ x[k-\beta_1] + \gamma x[k-\beta_2] \}$; $\alpha, \beta \in \mathbb{R}$, $\beta_1, \beta_2 \in \mathbb{Z}$, $\beta_2 > \beta_1$.

a) $y[n] = \sum_{k=-\infty}^{n-1} \alpha^{n-k} x[k-\beta_1] + \sum_{k=-\infty}^{n-1} \alpha^{n-k} \gamma x[n-\beta_2]$

$y[n] = \sum_{k=-\infty}^{+\infty} \alpha^{n-k} x[k-\beta_1] u[k+(n-1)] + \sum_{k=-\infty}^{+\infty} \alpha^{n-k} \gamma x[n-\beta_2] u[-k+(n-1)]$

$y[n] = x[-\beta_1] * \alpha^n u[n-1] + \gamma x[n-\beta_2] * \alpha^n u[n-1]$

$y[n] = x[n] * \delta[n-\beta_1] * \alpha^n u[n-1] + \gamma x[n] * \delta[n-\beta_2] * \alpha^n u[n-1]$

$y[n] = x[n] * \alpha^{n-\beta_1} u[n-(\beta_1+1)] + x[n] * \gamma \alpha^{n-\beta_2} u[n-(\beta_2+1)]$

$y[n] = x[n] * \{ \alpha^{n-\beta_1} u[n-(\beta_1+1)] + \gamma \alpha^{n-\beta_2} u[n-(\beta_2+1)] \}$

$y[n] = x[n] * h[n]$

$\Rightarrow h[n] = \alpha^{n-\beta_1} u[n-(\beta_1+1)] + \gamma \alpha^{n-\beta_2} u[n-(\beta_2+1)]$

$h[n] = \alpha^{n-\beta_1+1-1} u[n-(\beta_1+1)] + \gamma \alpha^{n-\beta_2+1-1} u[n-(\beta_2+1)]$

$h[n] = \alpha \alpha^{n-(\beta_1+1)} u[n-(\beta_1+1)] + \gamma \alpha \alpha^{n-(\beta_2+1)} u[n-(\beta_2+1)]$

$\alpha^n u[n] \longrightarrow \frac{1}{1-\alpha z^{-1}} \quad |z| > |\alpha|$

$x[n-n_0] \longrightarrow z^{-n_0} X(z)$

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$$H(z) = \alpha \left\{ \frac{z^{-(\beta_1+1)}}{1-\alpha z^{-1}} + \frac{\gamma z^{-(\beta_2+1)}}{1-\alpha z^{-1}} \right\}$$

$$H(z) = \alpha \left\{ \frac{z^{-(\beta_1+1)} + \gamma z^{-(\beta_2+1)}}{1-\alpha z^{-1}} \right\}$$

$$H(z) = \alpha \left\{ \frac{\frac{z^{\beta_2+1} + \gamma z^{\beta_1+1}}{z^{\beta_1+1}\, z^{\beta_2+1}}}{\frac{z-\alpha}{z}} \right\}$$

$$H(z) = \alpha \left\{ \frac{z^{\beta_2+1} + \gamma z^{\beta_1+1}}{z-\alpha} \cdot \frac{z}{z^{\beta_1+1} \cdot z^{\beta_2+1}} \right\}$$

$$H(z) = \alpha \left\{ \frac{z^{\beta_2} + \gamma z^{\beta_1}}{z-\alpha} \cdot \frac{z^2}{z^{\beta_1+\beta_2} z^2} \right\}$$

$$H(z) = \alpha \left\{ \frac{z^{\beta_1}\left(z^{\beta_2-\beta_1} + \gamma\right)}{(z-\alpha) z^{\beta_1} z^{\beta_2}} \right\}$$

$$H(z) = \frac{\alpha\left(z^{\beta_2-\beta_1} + \gamma\right)}{(z-\alpha) z^{\beta_2}}$$

ES CAUSAL $\Rightarrow$ $\beta_1 + 1 \geq 0 \wedge \beta_2 + 1 \geq 0 \wedge \beta_2 > \beta_1$

$\beta_1 \geq -1 \wedge \beta_2 \geq -1 \wedge \beta_2 > \beta_1$

CEROS: •) $z^{\beta_2-\beta_1} = -\gamma \rightarrow \boxed{z = \sqrt[\beta_2-\beta_1]{-\gamma}}$

POLOS: •) EN $z=0$ DE ORDEN β_2

•) EN $z=\alpha$

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b) PARA QUE SEA ESTABLE $|\alpha| < 1$; $\forall \alpha \in \mathbb{R}$.

c) PARA SER DE FASE MÍNIMA SU INVERSA TAMBIÉN TIENE QUE SER ESTABLE Y CAUSAL

$$H(z) \begin{cases} \rightarrow C_k & \text{CEROS} \\ \rightarrow P_k & \text{POLOS} \end{cases}$$

PARA QUE $H^{-1}(z)$ SEA CAUSAL: $|z| > \max|C_k|$

" " " " ESTABLE: $\max|C_k| < 1$

$\Rightarrow$ NECESITO QUE LOS POLOS Y LOS CEROS ESTÉN EN EL INTERIOR DE LA CIRCUNFERENCIA UNIDAD.

NO HABÍA RESTRICCIÓN PARA α.

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